

$\dot{\varphi}(t) = \emptyset$, light
max das krus

$$\dot{\varphi}(t) = \omega$$

ADD external light

$$\begin{aligned}\dot{\varphi} &= \omega - A \sin(\varphi(t) - \psi(t)) \\ \dot{\psi} &= \Omega \\ \underbrace{\dot{\varphi} - \dot{\psi}}_{\Theta} &= \omega - \Omega - A \sin(\underbrace{\varphi - \psi}_{\Theta})\end{aligned}$$

external light

$$\dot{\Theta} = \omega - \Omega - A \sin \Theta$$

2-

$$\dot{\varphi}_i = \omega - A \sin(\varphi_i - \psi)$$

$$\dot{\psi} = \Omega$$

$$\dot{\theta}_i = \omega - \Omega - A \sin \theta_i$$

$$\dot{\theta}_i = \emptyset$$

$$\dot{\theta} = \omega - \Omega - A \sin \theta$$

$$t = T \tau, [t] = [T] = \text{Time}$$

$$[\tau] = 1$$

$$\frac{d}{dt} = \frac{1}{T} \frac{d}{d\tau}$$

3-

$$\frac{d}{dT} \theta(T) = \omega - \Omega - A \sin \theta(T)$$

$$\frac{d}{dT} \theta(T) = (\omega - \Omega)T - AT \sin \theta(T)$$

$$AT = 1; T = 1/A$$

$$\frac{d}{dT} \theta(T) = \frac{\overset{\text{internal}}{\omega} - \overset{\text{external}}{\Omega}}{A} - \sin \theta(T)$$

$$\left| \frac{-\omega + \Omega}{A} \right| < 1$$

$$-A < \Omega - \omega < A$$

$$-A + \omega < \Omega < A + \omega$$

4-

$$\dot{\theta} = f(\theta)$$

$$\theta \equiv \text{mod}(\theta, 2\pi)$$

$$f(\theta + 2\pi) = f(\theta)$$

fixed point stable

$$\text{if } \frac{\partial f}{\partial \theta} < 0$$

unstable, if $\frac{\partial f}{\partial \theta} > 0$

$$\dot{\theta}_i = \omega_i + \sum_{jk} A_{ijk} \sin(\theta_i - \theta_j - \theta_k)$$

frequency is a
function of i

6

$$\text{Let } \underline{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{n1} & \dots & \dots & a_{nn} \end{bmatrix}$$

$$\frac{d}{dt} \underline{x}(t) = \underline{A} \underline{x}(t);$$

$$\ddot{x}(t) + \omega^2 x(t) = 0$$

$$\dot{x} = y$$

$$\dot{y} = -\omega^2 x \Rightarrow \dot{y} = \ddot{x} = -\omega^2 x$$

$$\frac{1}{T^2} \frac{d^2}{d\tau^2} x(\tau) + \omega^2 x$$

$$\frac{d^2}{d\tau^2} x(\tau) + T^2 \omega^2 x(\tau) = 0$$

$$T = 1/\omega$$

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -x \end{aligned} ; \quad \underline{r} = \begin{pmatrix} x \\ y \end{pmatrix} ; \quad \underline{v} = \dot{\underline{r}} = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}$$

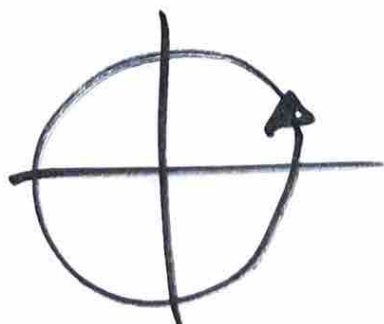
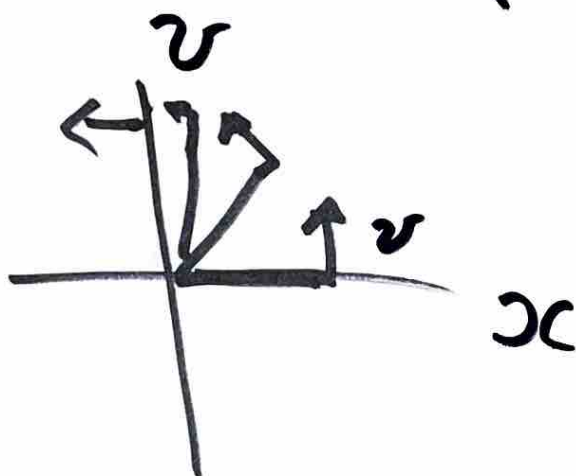
$$\underline{v} = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

8 =

$$\underline{r} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\underline{v} = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

$$\underline{r}^T \cdot \underline{v} = (x, y) \begin{pmatrix} y \\ -x \end{pmatrix} = xy - yx = 0$$



-9-

Section

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$\searrow \rightarrow$ vector
 $\searrow \rightarrow$ matrix

$$\dot{\underline{x}} = \epsilon \underline{A} \underline{x}$$

$$0 < \epsilon < 1$$

$$\underline{x}(t=0) =$$

$\underline{y}$

$$\underline{x}(t) = \underline{x}_0(t) + \epsilon \underline{x}_1(t) + \epsilon^2 \underline{x}_2(t) + \epsilon^3 \underline{x}_3(t) + \dots$$

-10-

$$\dot{x}_0 + \epsilon \dot{x}_1 + \epsilon^2 \dot{x}_2 + \epsilon^3 \dot{x}_3 + \dots$$

$$= A \cdot \epsilon (x_0 + \epsilon x_1 + \epsilon^2 x_2 + \epsilon^3 x_3 + \dots)$$

$$\underline{x}(t=0) = \underline{y} = x_{\text{initial}}$$

order 0

$$\underline{\dot{x}}_0 = 0$$

$$\underline{x}_0 = \underline{x}_0(t=0) = \underline{y} = x_{\text{initial}}$$

order

order 1

$$\epsilon \underline{\dot{x}}_1 = \epsilon \underline{A} \underline{x}_0 = \epsilon \underline{A} \underline{y}$$

$$\underline{\dot{x}}_1 = \underline{A} \underline{y};$$

$$\underline{x}_1(t) = \underline{A} \underline{y} \cdot t + \dots \quad \subseteq$$

$$\underline{x}(t=0) = \underline{y} = \overbrace{x_0(t=0)}^{\underline{y}} + \epsilon x_1(t=0) + \epsilon^2 x_2(t=0) + \dots$$

$$\underline{y} = \underline{x}(t=0) = \underline{y} + \epsilon \underline{x}_1(t=0) + \epsilon^2 \underline{x}_2(t=0) + \epsilon^3 \underline{x}_3(t=0) + \dots$$

$$\emptyset = \underline{x}_1(t=0) = \underline{x}_2(t=0) = \underline{x}_3(t=0) = \dots$$

Order 1: $\underline{x}_1(t) = \underline{A} \underline{y} t + \underline{C}, \underline{C} = \emptyset$

$$\underline{x}_1(t) = \underline{A} \underline{y} t$$

Order 2

$$\epsilon^2 \dot{\underline{x}}_2 = \epsilon^2 \underline{A} \underline{x}_1 = \epsilon^2 \underline{A} \cdot \underline{A} \underline{y} t$$

$$\underline{x}_2(t) = \underline{A}^2 \underline{y} t^2 / 2 + \underline{C} = \epsilon^2 \underline{A}^2 \underline{y} t$$

Initial condition

$$\underline{C} = \emptyset \quad \underline{x}_2 = \underline{A}^2 \underline{y} t^2 / 2$$

-12-

order 3

$$\dot{\underline{x}}_3 = \underline{A} \underline{x}_2 = \underline{A} \cdot \underline{A}^2 y t^2/2$$

$$\underline{x}_3 = \underline{A}^3 y \cdot t^3 / (2 \cdot 3)$$

$$\underline{x}(t) = y + \epsilon A y t + \epsilon^2 A^2 y t^2/2 + \epsilon^3 A^3 y t^3/6$$

-13-

$$\dot{\underline{x}}(t) = \underline{A} \underline{x}(t)$$

$$\underline{x}(t) = \sum_{k=0}^{\infty} \epsilon^k \underline{x}_k(t)$$

$$\sum_{k=0}^{\infty} \epsilon^k \dot{\underline{x}}_k = \sum_{k=0}^{\infty} \epsilon^{k+1} \underline{A} \underline{x}_k(t)$$

OR DER K $\dot{\underline{x}}_k = \underline{A} \underline{x}_{k-1}$

$$\underline{x}_0(t=0) = \underline{y}$$

$$\underline{x}_k = \underline{A}^k \cdot \underline{y} \cdot t^k / k!$$

$$\underline{x}_0 = \underline{y} \checkmark$$

$$\dot{\underline{x}}_{k-1} = \underline{A} \underline{x}_{k-1} \Rightarrow \dot{\underline{x}}_{k-1} = \underline{A}^{k-1} \underline{y} \frac{t^{k-1}}{(k-1)!}$$

$$\underline{x}_{k+1} = \underline{A}^{k+1} \underline{y} \frac{t^{k+1}}{(k+1)!}$$

-14-

$$e^x \stackrel{\text{definition}}{=} \sum_{k=0}^{\infty} x^k / k!$$

$$\frac{d}{dx} e^x = e^x$$

$$e^{\underline{A}} \stackrel{\text{def}}{=} \sum_{k=0}^{\infty} \underline{A}^k / k!$$

IVP

$$\underline{\dot{x}} = \underline{A} \underline{x}$$

$$\underline{x}(t=0) = \underline{y}$$

$$\Rightarrow \underline{x}(t) = \sum_{k=0}^{\infty} \frac{\underline{A}^k \underline{y} t^k}{k!}$$

$$\underline{x}(t) = e^{\underline{A}t} \underline{y}$$

-15-

$$\dot{\underline{x}} = \epsilon \underline{A} \underline{x}, \quad \underline{x}(t=0) = \underline{y}$$

$$\dot{\underline{x}}_{k+1} = \epsilon \underline{A} \underline{x}_k$$

$$\underline{x}(t=0) = \underline{y}_0$$

$$\underline{x}(t) = \sum_{k=0}^{\infty} \epsilon^k \underline{A}^k \frac{t^k}{k!} \underline{y}$$

Perturbation theory

$$0 < \epsilon \ll 1$$

-15-

Let A be $n \times n$ matrix

Eigen values $\lambda_1 \dots \lambda_n$ eigenvectors
 $x_1 \dots x_n$

$$P = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{n1} \\ x_{12} & x_{22} & & \vdots \\ x_{13} & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ x_{1n} & x_{2n} & & x_{nn} \end{pmatrix}$$

$\uparrow$
eigenvector
1
of A

$\rightarrow$ eigenvector
 2

$$A = P D P^{-1}$$

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

16

$$\|\underline{x}\|_m = \left(\sum_{k=1}^n |x_k|^m \right)^{1/m}$$

-16-

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{x}(t=0) = \underline{y}$$

$$\underline{x}(t) = \sum_{k=0}^{\infty} \underline{A}^k \frac{t^k}{k!} \underline{y}$$

$$\underline{x}(t) = \underline{y} + \sum_{k=1}^{\infty} \underline{A}^k \frac{t^k}{k!} \underline{y}$$

$$\frac{d}{dt} \underline{x}(t) = \sum_{k=1}^{\infty} \underline{A}^k \frac{t^{k-1}}{(k-1)!} \underline{y}$$

$$= \underline{A} \sum_{k=1}^{\infty} \underline{A}^{k-1} \frac{t^{k-1}}{(k-1)!} \underline{y} =$$

$$k = m+1$$

$$= \underline{A} \sum_{m=0}^{\infty} \underline{A}^m \frac{t^m}{m!} \underline{y}$$

$$= \underline{A} \underline{x}$$

17.

$$\begin{cases} \dot{\underline{x}} = \underline{A} \underline{x} \\ \underline{x}(t=0) = \underline{y} \end{cases}$$

$$\underline{x}(t) = \sum_{k=0}^{\infty} \underline{A}^k t^k / k! \underline{y} \equiv e^{\underline{A}t} \underline{y}$$

$$\underline{A} = \underline{P} \underline{D} \underline{P}^{-1}$$

$$\underline{x}(t) = \sum_{i=1}^n C_i e^{\lambda_i t} \underline{x}_i$$

$\underline{P}$ - eigenvectors
 $\underline{D}$ - diagonal matrix